A2 Q5

Tuesday, October 13, 2026 5:30 PM

5. a) First, we set up the normal equation for this problem.

$$S(\beta) = \sum_{i=1}^{n} [Y_i - \beta x_i]^2$$

$$\frac{\partial S}{\partial \beta} = -2 \sum_{i=1}^{n} [Y_i - \beta x_i] x_i$$

Set this derivative equal to zero, solve for β

$$0 = -2 \sum_{i=1}^{n} [Y_i - \beta x_i] x_i \quad \text{(divide by } -2\text{)}$$

$$\Rightarrow 0 = \sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} \beta x_i^2 \quad \text{(pull out constant } \beta\text{)}$$

$$\Rightarrow \beta \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} Y_i x_i$$

$$\Rightarrow \beta = \left(\sum_{i=1}^{n} Y_i x_i\right) / \left(\sum_{i=1}^{n} x_i^2\right)$$

$$= \hat{\beta}$$

Where $\hat{\beta}$ is the least squares estimator of β

b) Similar to the assumptions in the slides, let all Y_i be independent and $\text{Var}(Y_i) = \sigma^2$ (common variance), and $E[Y_i] = \beta x_i$ for the linear model in a). We treat x_i as a constant.

Thus:

$$E[\hat{\beta}] = \left(\sum_{i=1}^{n} x_i E[Y_i]\right) / \sum_{i=1}^{n} x_i^2$$

$$= \sum_{i=1}^{n} \beta x_i^2 / \sum_{i=1}^{n} x_i^2 \quad (\beta \text{ is a constant})$$

$$= \beta$$

$\therefore E[\hat{\beta}]$ is unbiased

$$E[\tilde{\beta}] = \sum_{i=1}^{n} E[Y_i] / \sum_{i=1}^{n} x_i$$

$$= \sum_{i=1}^{n} \beta x_i / \sum_{i=1}^{n} x_i \quad (\beta \text{ is a constant})$$

$$= \beta$$

$\therefore E[\tilde{\beta}]$ is unbiased

Now for deriving the variance:

$$\text{Var}(\hat{\beta}) = \text{Var}\left(\sum_{i=1}^{n} x_i Y_i\right) / \text{Var}\left(\sum_{i=1}^{n} x_i^2\right)$$

$$= \sum_{i=1}^{n} x_i^2 \text{Var}(Y_i) / \left(\sum_{i=1}^{n} x_i^2\right)^2$$

$$= \sum_{i=1}^{n} x_i^2 \sigma^2 / \left(\sum_{i=1}^{n} x_i^2\right)^2$$

$$= \sigma^2 / \sum_{i=1}^{n} x_i^2$$

$$= \sigma^2 / n^2 \bar{x}^2 \quad \text{(property from slides)}$$

$$\text{Var}(\tilde{\beta}) = \text{Var}\left(\sum_{i=1}^{n} Y_i\right) / \text{Var}\left(\sum_{i=1}^{n} x_i\right)$$

$$= \sum_{i=1}^{n} \text{Var}(Y_i) / \sum_{i=1}^{n} x_i^2$$

$$= n\sigma^2 / \sum_{i=1}^{n} x_i^2$$

$$= n\sigma^2 / n^2 \bar{x}^2 \quad \text{(property from slides)}$$

$$= \sigma^2 / n\bar{x}^2$$

No need to use the hint, we may directly compare the variances:

$$\sigma^2 / n^2 \bar{x}^2 \leq \sigma^2 / n\bar{x}^2$$

$$\Rightarrow \text{Var}(\hat{\beta}) \leq \text{Var}(\tilde{\beta})$$

This is true as the denominator has n to a greater power in $\text{Var}(\hat{\beta})$, and the two variances may be equal when $n = 1$, or when x_i are equal, thus $x_i = \bar{x}\ \forall i$.

This is also intuitive because $\tilde{\beta}$ has less terms in its least squares estimator relative to $\hat{\beta}$; the more variables the higher the variance of a model typically.

c) No, we did not use distribution assumptions throughout this question at all.
We only used the assumptions that:
- $E[\varepsilon_i] = 0$ to show unbiasedness
- $Var(\sigma^2) = \sigma^2$ to then also assume $Var(Y_i) = \sigma^2$ by independence

This follows with the similar derivations present in the slides.